

Problem #1: Stellar Evolution (26 points)

We have officially covered all the top level things necessary to understand the general principles of stellar evolution! This problem asks you to prove it.

Problem #1a: Accessing and plotting stellar models (10 pts)

Since we don't have access to a supercomputer that can help us run many stellar evolution models, we'll be using the MESA Isochrones and Stellar Tracks (MIST) packaged grids. Your first task is to download the theoretical evolutionary tracks from the [MIST website](#) for nonrotating stars at solar metallicity ($[Fe/H] = 0.0$). Once you have the models, plot the evolution in the theoretical H-R Diagram of a $1 M_{\odot}$, $5 M_{\odot}$, $10 M_{\odot}$, and $20 M_{\odot}$ star on five separate figures. Be sure to label your axes and make sure that hotter temperatures start at the left of the x-axis!

Problem #1b: Accessing and plotting stellar models (16 pts)

For each star, label (either with text in the plotting software or by hand) the critical phases of evolution that are included in each evolutionary track. You should explain the reasoning for every slope and direction change that the star makes as it evolves across the H-R Diagram for each of the stars you have plotted.

Problem #2: Self Reflections in Research (9 pts)

Now that you have some distance from your project 1 presentations, it's time to reflect. Write down three things that worked well in your project and three things that didn't. For each of the three things that didn't work, write down a strategy for how you can mitigate running into those problems for project 2.